Customize settings with environment variables

By: **Arknox**

Email: info@arknox.in

Contact: +91-8787388949

This is the configuration reference for the actual Quizs source. Edit the backend's private .env file, not the documentation files or Flutter assets. Some values are controlled by administrator forms; others are server-only configuration. The tables show current defaults and the effect of changes.

Screen / location: Backend project → .env → config/*.php

Step by step

1. Identify the backend serving the app. On cPanel, open File Manager, enable Show Hidden Files, and locate .env beside artisan and composer.json in the Laravel project directory. Locally, use Quiz Application Admin/.env. Do not confuse it with .env.example.

2. Make a private backup of .env before editing. Use .env.example only as a template for a new installation. Do not overwrite a live .env or replace its APP_KEY to change a feature.

3. Find the exact variable below. Replace its existing line, or add one line if absent. Keep one definition per key. Variables such as XP_PER_LEVEL are supported by config/quiz.php even though .env.example does not list them.

4. Use KEY=value syntax. Use true/false for booleans, plain integers for counts, and quotes around values containing spaces or a # character. For example OFFERWALL_PLACEMENT="#WebOfferwall". Preserve JSON quoting and private-key formatting; the verified Firebase admin form is safer than manually pasting JSON.

5. Save, then run php artisan config:clear from the backend directory. For a production deployment that uses cached configuration, run php artisan config:cache afterward. Restart long-running queue workers with php artisan queue:restart, or restart the relevant managed process so it loads the new configuration.

6. Reload the relevant administrator page and app flow. Sign in again when checking a changed heartbeat interval. Compare an actual result or delivery event, rather than relying only on a Configured badge.

7. If a change fails, restore the private backup, rebuild configuration cache, restart workers, and inspect storage/logs/laravel.log. Keep passwords and service-account keys out of public screenshots and this guide.

Field definitions and change impact

Field / Feature	Definition and source	Effect of changes
APP_NAME / APP_URL	Application name and backend URL. .env.example uses Laravel and http://localhost.	Use your product name and final HTTPS domain. URL changes affect generated links/artwork; Flutter ApiClient.baseUrl must be changed separately and rebuilt.
APP_ENV / APP_DEBUG	Template: local / true. Production: production / false.	Controls environment and debug output, not app content or colors.
APP_KEY	Installation-specific encryption key.	Retain the existing key on upgrades. Generating another key is not a customization step.
APP_TIMEZONE / BUSINESS_TIMEZONE	Both default to Asia/Kolkata. Business timezone falls back to APP_TIMEZONE.	BUSINESS_TIMEZONE controls today/month boundaries in targets and statistics. Changing it can change which date activity belongs to; it does not rewrite historical daily rows.
APP_LOCALE / APP_FALLBACK_LOCALE	Template: en / en; Laravel application locales.	Do not control Flutter interface language or bilingual question fallback. Those have separate settings.
DB_CONNECTION / DB_HOST / DB_PORT / DB_DATABASE / DB_USERNAME / DB_PASSWORD	Template uses sqlite; hosted MySQL needs its own connection values.	Selects the application database. Changing credentials does not migrate old records into another database.
DEFAULT_DAILY_QUIZ_TARGET	Default 20; fallback in config/quiz.php.	Used only if no global target has been saved in app_settings and the user has no custom target. It does not override the General Settings value.
XP_PER_COMPLETED_DAILY_TARGET	Default 10; integer reward in config/quiz.php.	Changes XP awarded for future first daily-target completions. Does not retroactively add XP to previously completed days. Use a nonnegative product-approved value.
XP_PER_LEVEL	Default 100; LevelService uses at least 1.	Computed level = floor(total XP ÷ XP_PER_LEVEL) + 1. Changing the curve can alter derived levels without changing earned XP; stored current_level used by ranking is not bulk-recalculated by editing .env.
HEARTBEAT_INTERVAL_SECONDS	Default 45; sent in authentication responses.	Changes the interval the app receives at login. Existing app state may retain its old interval until sign-in/reinitialization.

Field / Feature	Definition and source	Effect of changes
ONLINE_TIMEOUT_SECONDS	Default 120.	Controls when last_active_at is considered online. Keep it longer than the expected heartbeat interval to avoid intermittent offline estimates.
ACTIVE_USERS_WINDOW_DAYS	Default 30.	Rolling active-user count window; not the separate Active This Month calendar count.
PERFORMANCE_EXCELLENT_THRESHOLD	Default 90; percent correct out of all quiz questions.	Selects excellent result classification; not the per-quiz passing threshold.
PERFORMANCE_GOOD_THRESHOLD	Default 70.	Selects good classification below excellent.
PERFORMANCE_AVERAGE_THRESHOLD	Default 50; below this is low.	Keep thresholds ordered excellent ≥ good ≥ average within 0-100. Display wording is in Flutter results_screen.dart.
FALLBACK_LANGUAGE	Default en; supported question languages are en and hi.	Fallback translation for bilingual questions when requested content is missing. Does not translate text or change the app's interface-language preference.
MAIL_MAILER	Template/default log; Email Configuration Save sets smtp.	log writes mail to logs instead of sending to recipients. Set smtp for configured SMTP delivery.
MAIL_HOST / MAIL_PORT	Defaults 127.0.0.1 / 2525, not universal provider values.	Use the actual provider endpoint and port. Email Configuration writes these variables.
MAIL_USERNAME / MAIL_PASSWORD	Template null; private SMTP credentials.	Email Configuration writes username and a supplied replacement password. Blank password in the admin form retains the stored one.
MAIL_FROM_ADDRESS / MAIL_FROM_NAME	Template hello@example.com / APP_NAME.	Sender email/name for recipients. Configure a provider-authorized sender; the admin form writes these values.
MAIL_SCHEME / MAIL_URL / MAIL_EHLO_DOMAIN	Optional SMTP scheme, transport URL, and greeting domain from config/mail.php.	Advanced server-only configuration. Match the provider's settings. A populated MAIL_URL can supply transport parameters; resolve conflicts with individual fields. This source does not read MAIL_ENCRYPTION.
FIREBASE_API_KEY	Blank in the template; read by API authentication.	Firebase authentication configuration, separate from FCM push credentials and native google-services.json.

Field / Feature	Definition and source	Effect of changes
FCM_PROJECT_ID	Blank in the template.	Push target project. Firebase Verify & Save writes the project_id from the supplied service-account JSON.
FCM_CREDENTIALS_JSON	Blank in the template; complete service-account JSON inline.	Preferred when nonempty. The Firebase admin form verifies with Google before writing it. Invalid nonempty JSON does not fall back to FCM_CREDENTIALS_PATH.
FCM_CREDENTIALS_PATH	Blank; private server file path to service-account JSON.	Used only when inline JSON is empty. Keep the file outside public/. Manual editing does not perform the admin form's Google verification.
TURNSTILE_ENABLED	Configuration default true; may be absent from .env.example.	Enables administrator login verification. Login Captcha writes true/false here.
TURNSTILE_SITE_KEY / TURNSTILE_SECRET_KEY	Configuration defaults are public test keys.	Use your real domain's keys for production. The admin form writes site key and a supplied secret; blank replacement secret retains the previous one.
QUEUE_CONNECTION	Template database.	Queued mail needs queue processing and database queue tables. Changing the driver does not move existing queued jobs.
CRON_SECRET	Blank in the template.	Private shared key for GET /api/cron/run-queue. Required when using the protected HTTP queue runner; not a replacement for SMTP credentials.
SESSION_DRIVER / SESSION_LIFETIME	Template database / 120 minutes.	Administrator session storage and lifetime. Database sessions need the session table; not a quiz-duration setting.
CACHE_STORE	Template database.	Cache backend; database cache requires its tables. Changing the driver is infrastructure configuration, not target editing.
OFFERWALL_ENABLED	Default false.	Enables the offerwall launch feature. Requires provider settings as well.
OFFERWALL_SDK_KEY / OFFERWALL_PLACEMENT	Blank SDK key; default placement #WebOfferwall.	Provider launch configuration. Quote the placement so # is preserved.
OFFERWALL_CALLBACK_SECRET / OFFERWALL_CALLBACK_MODE	Private secret; default mode hmac_sha256.	Verifies provider callbacks. Completion records do not automatically credit wallet, quiz points, or XP.

Field / Feature	Definition and source	Effect of changes
FILESYSTEM_DISK / AWS_*	Template local, plus optional AWS storage fields.	Category uploads explicitly use the category_images disk. Changing the default disk alone does not move existing uploads or make that flow use S3. Configure the named disk separately if changing its storage.
LOG_CHANNEL / LOG_LEVEL	Template stack / debug.	Logging configuration. Does not change reports, notifications, or app strings. Protect server logs from public access.

Current implementation limit
All values are configuration guidance; this guide does not edit your running server. Keep credentials in the private backend .env and preserve existing application keys. Product defaults in the current source are starting values, not a promise of a finalized XP economy.

Data and behavior in detail

Which settings are saved in the database?

Settings → General stores app_settings.global_daily_quiz_target. Users → Edit stores custom_daily_target on the user record. Priority is individual custom target → saved global target → DEFAULT_DAILY_QUIZ_TARGET. The daily row then freezes that effective target. Example: .env fallback 5, saved global 20, and user override 8 means that user gets 8 for a new day; a global user gets 20.

Which admin forms write .env?

Email Configuration writes MAIL_MAILER=smtp, MAIL_HOST, MAIL_PORT, MAIL_USERNAME, MAIL_FROM_ADDRESS, MAIL_FROM_NAME, and a nonblank replacement MAIL_PASSWORD. Firebase writes FCM_PROJECT_ID and FCM_CREDENTIALS_JSON after verification. Login Captcha writes TURNSTILE_SITE_KEY, TURNSTILE_ENABLED, and a nonblank replacement TURNSTILE_SECRET_KEY. Admin save and manual .env editing change the same values; a later admin save can replace your manual edits.

Automatic backups and configuration refresh

Those three forms use EnvFileWriter: it backs up the previous nonempty .env in storage/app/private/env-backups, keeps up to 20 backups, writes the file, and runs config:clear. It does not automatically rebuild config cache or restart long-running workers. Manual File Manager editing does not create that automatic backup or clear the cache.

Example: change future XP and result classification

```
XP_PER_COMPLETED_DAILY_TARGET=15
XP_PER_LEVEL=150
PERFORMANCE_EXCELLENT_THRESHOLD=95
PERFORMANCE_GOOD_THRESHOLD=75
PERFORMANCE_AVERAGE_THRESHOLD=50
```

These are sample customization values, not new product defaults. Existing earned XP remains unchanged; inspect derived level and stored ranking level separately after a curve change. Test an attempt around each result threshold.

Example: configure SMTP using .env

```
MAIL_MAILER=smtp
MAIL_HOST=smtp.your-provider.example
MAIL_PORT=587
MAIL_USERNAME="YOUR_SMTP_USERNAME"
MAIL_PASSWORD="YOUR_PRIVATE_SMTP_PASSWORD"
MAIL_FROM_ADDRESS="no-reply@your-domain.example"
MAIL_FROM_NAME="Quizs"
```

Replace placeholders with provider values. Port 587 here is an example, not a requirement. Configure MAIL_SCHEME if your provider requires it; this project uses MAIL_SCHEME, not MAIL_ENCRYPTION. After cache refresh, use Email Configuration → Send Test and test queued OTP/account mail separately.

Example: use a private Firebase credentials file

```
FCM_PROJECT_ID="YOUR_FIREBASE_PROJECT_ID"
FCM_CREDENTIALS_JSON=
FCM_CREDENTIALS_PATH="/home/account/private/firebase-service-account.json"
```

Replace the example path with an existing readable private file containing valid JSON. Empty inline JSON permits the file fallback. Confirm that the native app uses the same project, registers a token, and has permission.

Apply changes from Terminal or SSH

```
php artisan config:clear
php artisan config:cache
php artisan queue:restart
```

Run from the backend directory containing artisan. config:cache is appropriate when production uses cached configuration. queue:restart asks persistent workers to restart after their current job; a process supervisor must start them again. A one-shot HTTP cron worker starts fresh for each invocation. If you have no terminal, arrange equivalent deployment commands through the host's supported tools; saving a manual .env edit alone does not invalidate an existing configuration cache.

What cannot be customized through these variables?

Category/question content is managed in its forms. Quiz language is chosen at creation. Per-quiz passing percentage belongs to the quiz record. Ranking order is code in RankingService/QuizRankingService. App API URL, logo, theme, strings, ad-unit IDs, and the fixed Next Trial countdown are Flutter/platform source changes requiring a new build. Backend .env does not automatically become Flutter configuration.

Historical effects and verification

New target defaults do not rewrite existing daily rows. New reward values do not retroactively adjust earned XP. Editing thresholds does not automatically resubmit or rescore historical attempts. Changing XP_PER_LEVEL recalculates some derived level displays, while ranking reads stored current_level; a deliberate data migration is needed to make historical stored levels consistent. Do not invent a migration or reset production records just to test a setting.

Verification after saving

After each change, check its exact outcome: new-day target, one future XP award, login heartbeat/online timeout, boundary-case result classification, actual mail, real-device push, or a verified offerwall callback. If only documentation text changed, no application setting changed.

Technical sources

Server paths are inside Quiz Application Admin; lib paths are inside Quiz Application.

.env.example

config/app.php

config/quiz.php

config/mail.php

config/services.php

config/offerwall.php

app/Support/EnvFileWriter.php

app/Services/TargetService.php

app/Services/LevelService.php

app/Services/ProgressionService.php

app/Services/RankingService.php

app/Services/FcmService.php

app/Http/Controllers/Admin/EmailSettingsController.php

app/Http/Controllers/Admin/FirebaseSettingsController.php

app/Http/Controllers/Admin/TurnstileSettingsController.php

lib/services/activity_service.dart

Conclusion Section

Use this guide to configure Quizs, verify the actual app behavior, and keep administrator content, server configuration, and native app changes in their correct places.

Administrator access

Open your own backend domain followed by /admin/login. Use the credentials created for that deployment. Accounts, passwords, and URLs from unrelated reference projects do not apply to Quizs.

Release verification

Complete a category, quiz, question, attempt, result, target/profile, notification, and attendance test. Confirm native integrations separately on the intended device and provider account.

Contact for support

Email: info@arknox.in
Support Center: +91-8787388949

Arknox